

INDIGO / VAJRA

AI Powered Threat Intelligence & Risk Analysis Dashboard

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1. Project Overview

INDIGO/VAJRA is a comprehensive cybersecurity dashboard providing real-time threat intelligence, ransomware monitoring, and risk analysis capabilities. Built with modern technologies to deliver a production-ready, pixel-perfect user experience with dark theme, blue highlights, and red threat colors. **Key Features:** • Real-time threat intelligence monitoring • Ransomware incident tracking and analysis • Live global attack visualization • Domain risk analysis and security assessment • Executive summary and risk counters • Multi-source threat data aggregation • PDF report generation • WebSocket real-time updates

Technology Stack

Component	Technology
Frontend Framework	Next.js 15 + React 19
Backend Framework	FastAPI (Python)
Database	PostgreSQL
Caching	Redis
Styling	TailwindCSS
State Management	Zustand
Charts	Recharts
Authentication	JWT
Real-time	WebSockets
Task Scheduling	APScheduler

2. System Architecture

The system follows a modern microservices-inspired architecture with clear separation of concerns:

- Frontend Layer:** • Next.js App Router for server-side rendering • React components for UI rendering • Zustand for client-side state management • Axios for API communication
- Backend Layer:** • FastAPI for RESTful API endpoints • SQLAlchemy ORM for database operations • Redis for caching and session management • WebSocket manager for real-time updates
- Data Layer:** • PostgreSQL for persistent data storage • Redis for high-performance caching • File system for generated reports
- Integration Layer:** • Third-party API services (AlienVault, VirusTotal, etc.) • Background task scheduler • WebSocket broadcast system

3. Backend Documentation

Backend Structure: • FastAPI application with modular router architecture • JWT-based authentication with refresh tokens • PostgreSQL database with SQLAlchemy ORM • Redis caching layer for performance optimization • WebSocket support for real-time updates • APScheduler for background task management • Comprehensive error handling and logging

Key Backend Components: • main.py - Application entry point and router configuration • routes/ - API endpoint definitions (auth, dashboard, threat, ransomware, etc.) • services/ - Business logic and third-party integrations • models/ - Database models and schema definitions • schemas/ - Pydantic schemas for request/response validation • middleware/ - Security middleware (rate limiting, CORS, etc.) • websocket/ - WebSocket connection management • scheduler/ - Background task scheduling

API Routes

Route	Description
/api/auth/*	Authentication endpoints
/api/dashboard/*	Dashboard data and statistics
/api/threat-intelligence/*	Threat intelligence data
/api/ransomware/*	Ransomware incident tracking
/api/news/*	Cyber threat news feed
/api/reports/*	PDF report generation
/api/domain-risk/*	Domain risk analysis
/api/companies/*	Company management
/ws/dashboard	WebSocket for real-time updates

4. Frontend Documentation

Frontend Structure: • Next.js 15 with App Router architecture • React 19 for component-based UI • TypeScript for type safety • TailwindCSS for utility-first styling • Framer Motion for animations • Lucide Icons for iconography • Recharts for data visualization • React Simple Maps for geographic data

Key Frontend Pages: • / - Main dashboard with threat overview • /login - Authentication page • /alerts - Critical alerts management • /companies - Company and domain analysis • /domain-analysis - Deep domain security analysis • /threat-intelligence - Threat intelligence dashboard • /ransomware - Ransomware incident tracking • /global-attacks - Global attack visualization • /executive-summary - Executive risk summary • /settings - Application settings

Key Components: • TopCards - Dashboard summary cards • ThreatMap - Interactive global attack map • AlertCards - Horizontal scrolling alerts • DomainDetails - Comprehensive domain analysis • Various chart components for data visualization

5. API Documentation

Authentication Endpoints: • POST /api/auth/register - Register new user • POST /api/auth/login - Login user • POST /api/auth/refresh - Refresh access token • POST /api/auth/logout - Logout user • GET /api/auth/profile - Get user profile **Dashboard Endpoints:** • GET /api/dashboard/summary - Get dashboard summary • GET /api/dashboard/alerts - Get critical alerts • GET /api/dashboard/attack-map - Get attack map data **Threat Intelligence Endpoints:** • GET /api/threat-intelligence - Get threat intelligence data • GET /api/threat-intelligence/trend - Get threat trend data • GET /api/threat-intelligence/actors - Get threat actors data **Ransomware Endpoints:** • GET /api/ransomware - Get ransomware incidents • GET /api/ransomware/stats - Get ransomware statistics **Report Endpoints:** • GET /api/reports/threat - Generate threat intelligence report (PDF) • GET /api/reports/ransomware - Generate ransomware report (PDF) • GET /api/reports/executive - Generate executive summary report (PDF)

6. Services & Integrations

Third-Party Integrations: • AlienVault OTX - Threat intelligence data • AbuseIPDB - IP reputation and abuse reporting • VirusTotal - Malware and URL analysis • ThreatFox - Threat intelligence indicators • Shodan - Internet-connected device search • NVD (National Vulnerability Database) - CVE data • Gridinsoft - Malware analysis • SSL Labs - SSL/TLS certificate analysis • URLScan.io - URL scanning and analysis

Internal Services: • PDF Service - Report generation with ReportLab • Dashboard Service - Dashboard data aggregation • Threat Service - Threat intelligence processing • Ransomware Service - Ransomware data management • Domain Analysis Service - Domain security analysis • WebSocket Manager - Real-time connection management • Activity Logger - User activity tracking

Background Tasks

Task	Frequency	Description
Ransomware Data Fetch	Every 15 minutes	Fetch ransomware incidents from threat-ransomware.live
Threat Intelligence Refresh	Every 30 minutes	Update threat indicators from AlienVault OTX
Dashboard Stats Refresh	Every 5 minutes	Refresh dashboard statistics
Activity Logging	Continuous	Track user activities and system events

7. Database Schema

Database Tables: • users - User accounts and authentication data • alerts - Critical security alerts and notifications • threat_feeds - Threat intelligence feed sources • attack_map - Geographic attack data for visualization • attack_events - Individual attack event records • threat_scores - Calculated threat scores and metrics • ransomware_groups - Known ransomware group information • ransomware_incidents - Ransomware attack incident records • countries - Country data for attack mapping • news - Cyber threat news articles • reports - Generated report metadata • activity_logs - User activity tracking logs • companies - Monitored companies and domains • domain_analysis - Domain analysis results

Database Technologies: • PostgreSQL 15+ - Primary relational database • Redis 7+ - Caching layer and session storage • SQLAlchemy ORM - Database interaction layer

8. Deployment Guide

Docker Deployment: The project includes Docker support for containerized deployment:

1. Build and start all services: `docker-compose up -d --build` 2. Access the application: •

Frontend: `http://localhost:3000` • Backend API: `http://localhost:8000` • API Documentation:

`http://localhost:8000/docs` **Local Development:** Backend Setup: • `cd backend` • `python -m`

`venv venv` • `source venv/bin/activate` • `pip install -r requirements.txt` • `uvicorn main:app`

`--reload` Frontend Setup: • `cd frontend` • `npm install` • `npm run dev` **Environment**

Configuration: Required environment variables are defined in `.env.example` files for both backend and frontend.

9. Security Features

Authentication & Authorization: • JWT-based authentication with access and refresh tokens • Password hashing with bcrypt • Token expiration and refresh mechanism • User profile management **API Security:** • Rate limiting with slowapi • CORS configuration for cross-origin requests • Security headers (CSP, XSS protection, etc.) • Input validation with Pydantic schemas • SQL injection prevention via ORM **Data Security:** • Encrypted password storage • Secure session management with Redis • Environment variable configuration for sensitive data • Database connection encryption **Monitoring & Logging:** • Comprehensive application logging • Activity logging for user actions • Error tracking and reporting • WebSocket connection monitoring

End of Project Documentation

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